

Sentinel Net-AI-Powered Network Intrusion Detection System (NIDS) using NSL-KDD Dataset

Title: Sentinel Net – AI- Powered Network Intrusion Detection System

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Abstract

The rise of cyber threats has highlighted the need for adaptive intrusion detection systems. This work presents *Sentinel Net*, a machine learning-based IDS developed using the NSL-KDD dataset. Preprocessing, hybrid resampling, and supervised classifiers such as Logistic Regression, Random Forest, and Decision Tree were employed. The system was evaluated using accuracy, recall, precision, F1-score, confusion matrices, and ROC-AUC curves. Results show that Random Forest and Decision Tree outperform Logistic Regression, especially in detecting minority attack classes.

Introduction

Intrusion Detection Systems (IDS) play a vital role in safeguarding networks from malicious activities. Traditional IDS rely on signatures or rules, which fail when faced with new or modified attack patterns. Machine learning provides a more robust approach by identifying anomalies from traffic features.

This project focuses on developing a data-driven IDS model using the NSL-KDD dataset. By integrating preprocessing, resampling, and advanced classifiers, the project ensures improved classification of both majority and minority attack categories.

Literature

Denning first introduced the idea of IDS in 1987, laying the groundwork for anomaly detection. The KDD Cup 1999 dataset was a benchmark but was later criticized for redundancy and imbalance. NSL-KDD addressed these issues, becoming a standard dataset for IDS research.

Over time, researchers have applied SVMs, Decision Trees, and Random Forests to intrusion detection, achieving strong performance. More recently, deep learning methods such as CNNs and RNNs have been explored. However, the challenge of detecting rare classes still

persists. This project builds on existing work by combining hybrid sampling with classical models for balanced detection.

Methodology

a. Database

The NSL-KDD dataset consists of 125,972 training and 22,543 testing records. Each record has 41 features grouped into basic, content, time-based, and host-based categories. The target classes include both normal traffic and 21 attack types such as Neptune, Smurf, Satan, and Buffer Overflow.

Link: [NSL-KDD](#)

Visualizations:

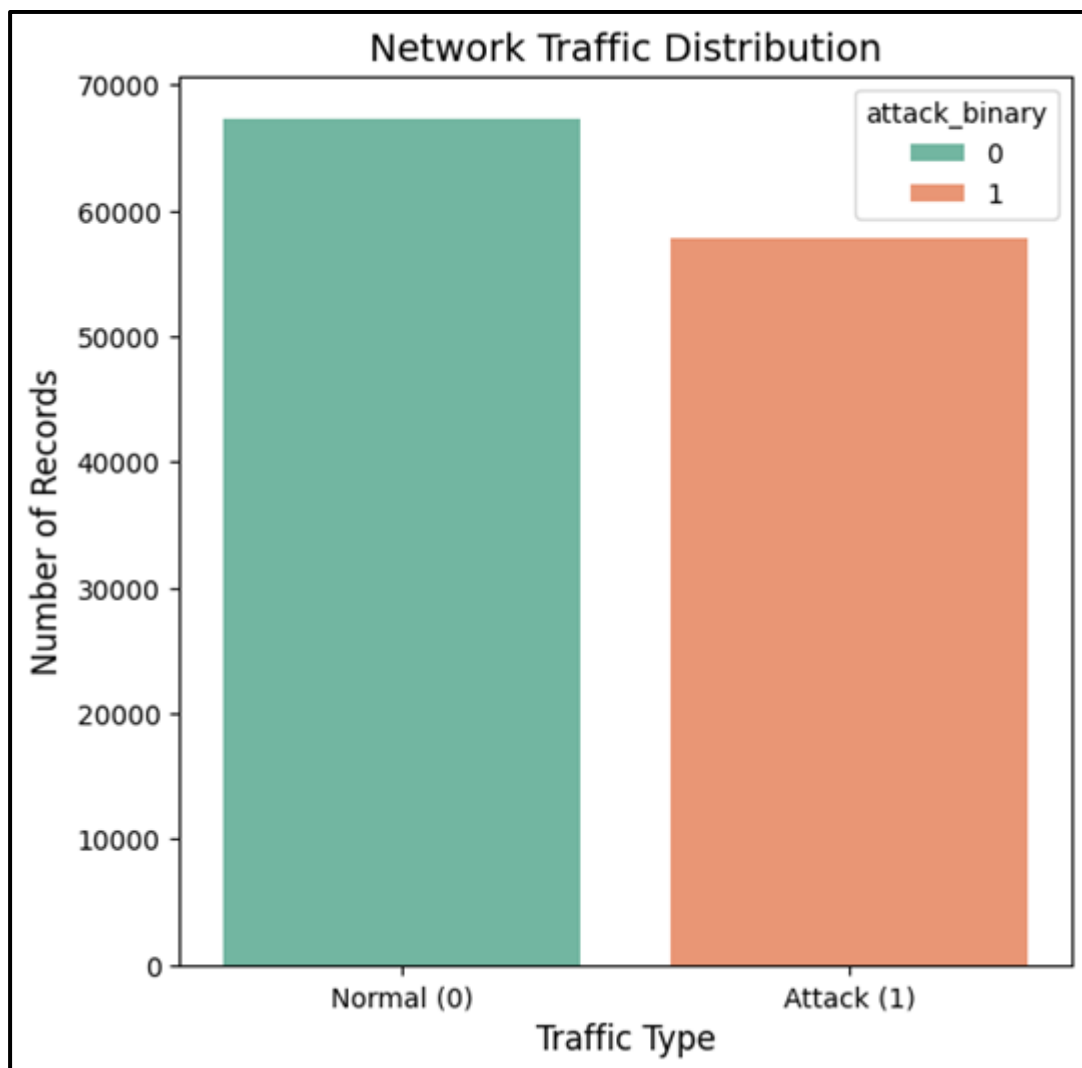


Figure 1: Network traffic distribution (binary classification) – Normal vs Attack.

b. Preprocessing

Preprocessing included handling categorical values, normalizing numerical features, and cleaning redundant entries. Severe imbalance in class distribution was addressed using the SMOTE-Tomek technique, which creates synthetic minority samples and removes borderline noise points for improved balance.

c. Novelty

The novelty lies in the integration of hybrid resampling with classical classifiers. Most studies rely solely on accuracy, but this project emphasizes macro F1-score to evaluate minority class detection. By focusing on fairness across classes, the approach ensures robustness against imbalance.

d. Proposed Method

Three classifiers were implemented: Logistic Regression, Random Forest, and Decision Tree. After resampling, the models were trained and tested on NSL-KDD. Their performances were evaluated using confusion matrices, classification reports, and ROC-AUC curves. The emphasis was on identifying which model best handled both common and rare attack types.

Results

EDA

Exploratory Data Analysis showed that normal and Neptune dominate the dataset. Minority classes such as Buffer Overflow, Land, and Guess Password had extremely low samples, highlighting imbalance. Feature correlations revealed redundancy, while statistical summaries indicated distinct separation between attack and normal traffic in some features.

Visualizations:

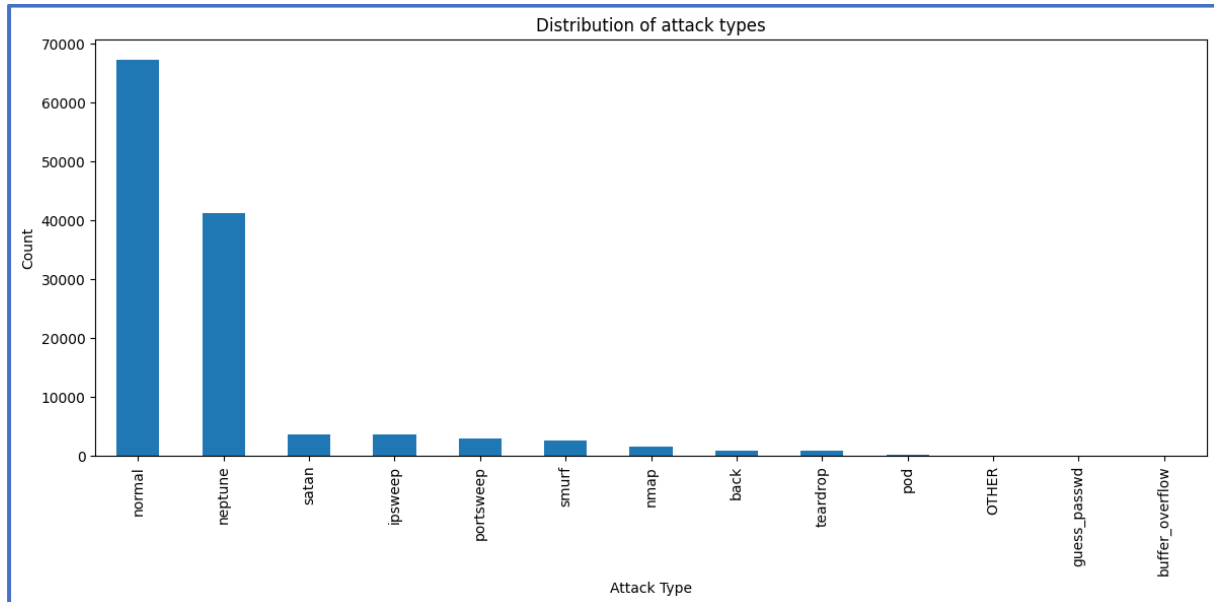


Figure 2: Count plots confirmed the imbalance between attack classes in Training Data

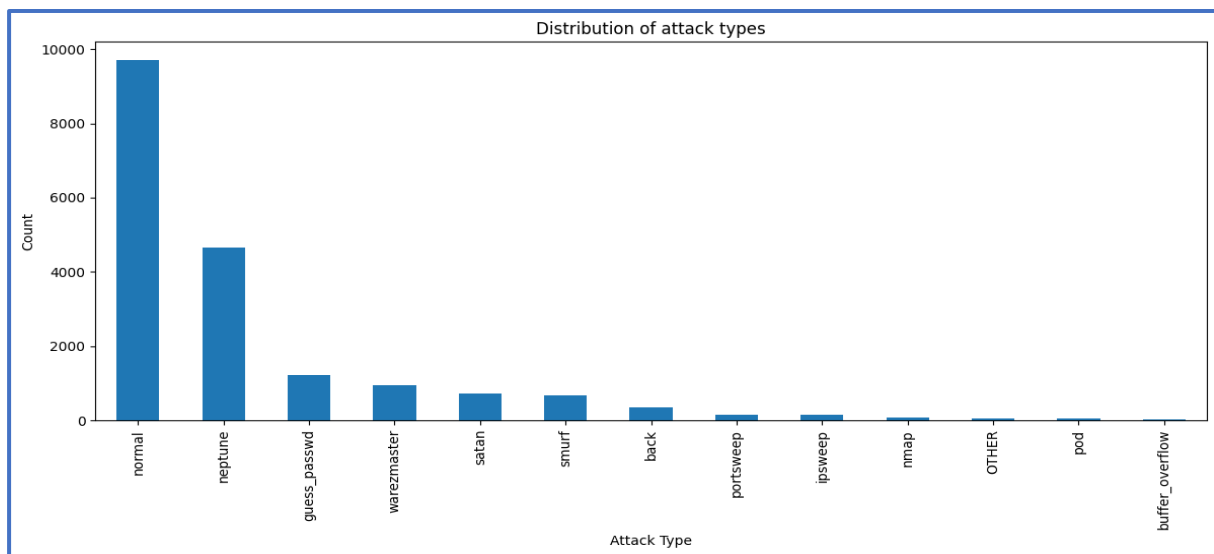


Figure 3: Count plots confirmed the imbalance between attack classes Testing Data.

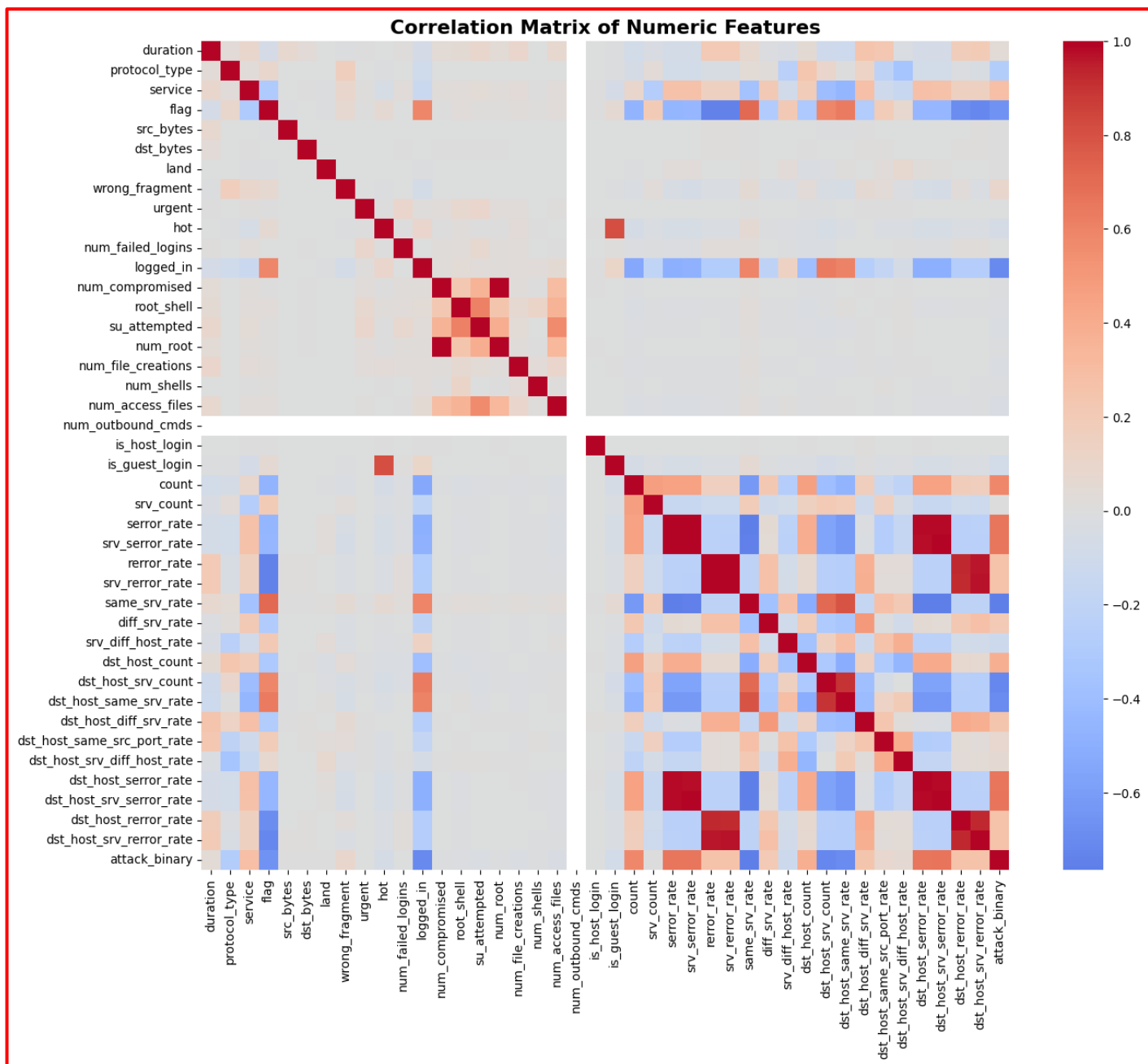


Figure 4: Heatmaps of correlations identified key features influencing attack detection.

Confusion Matrices

The confusion matrices highlighted differences in classifier performance. Logistic Regression misclassified several minority attacks, whereas Random Forest and Decision Tree provided stronger separation between normal and attack categories. This reinforced the effectiveness of tree-based models.

Visualizations:

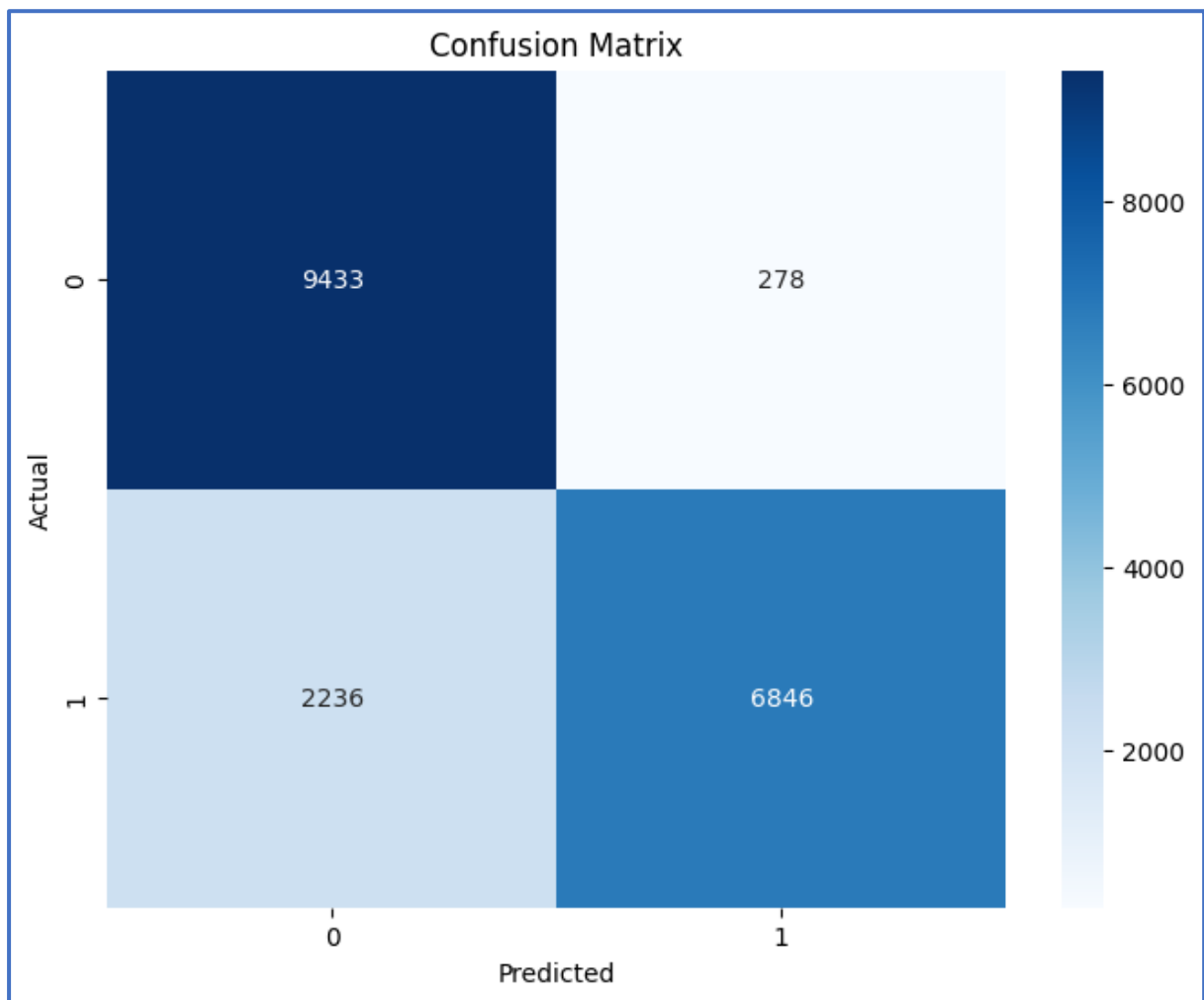


Figure 5: Confusion matrix for binary classification.

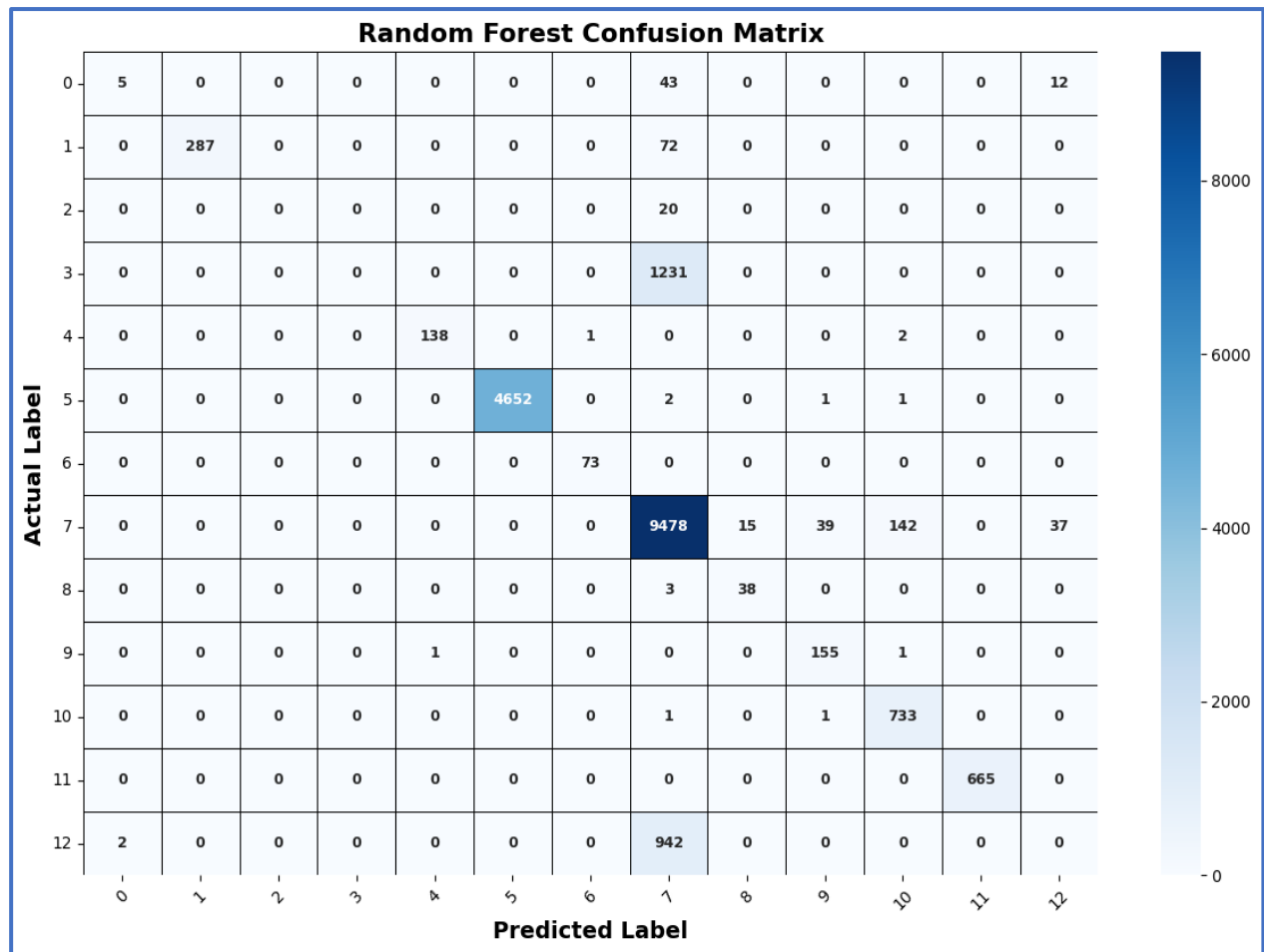


Figure 6: Confusion matrix for multiclass classification.

Results – Accuracy, Recall, Precision, F1-score

Logistic Regression achieved 97.11% accuracy on validation and 84.60% on test data.

Random Forest reached 99.91% validation accuracy and 86.62% on test data. Decision Tree achieved 99.82% validation accuracy and 87.20% test accuracy. Macro F1-scores demonstrated that Random Forest and Decision Tree handled minority classes better.

Model	Validation Accuracy	Test Accuracy	Macro F1-score For test accuracy
Logistic Regression	97.11%	84.60%	0.84
Random Forest	99.91%	86.62%	0.86
Decision Tree	99.82%	87.20%	0.87

Table 1: Model performance comparison table for Binary class classification

Model	Validation Accuracy	Test Accuracy	Macro F1-score For test accuracy
Logistic Regression	97.07%	84.22%	0.54
Random Forest	99.85%	86.33%	0.65
Decision Tree	99.84%	82.90%	0.59

Table 2: Model performance comparison table for Multiclass classification

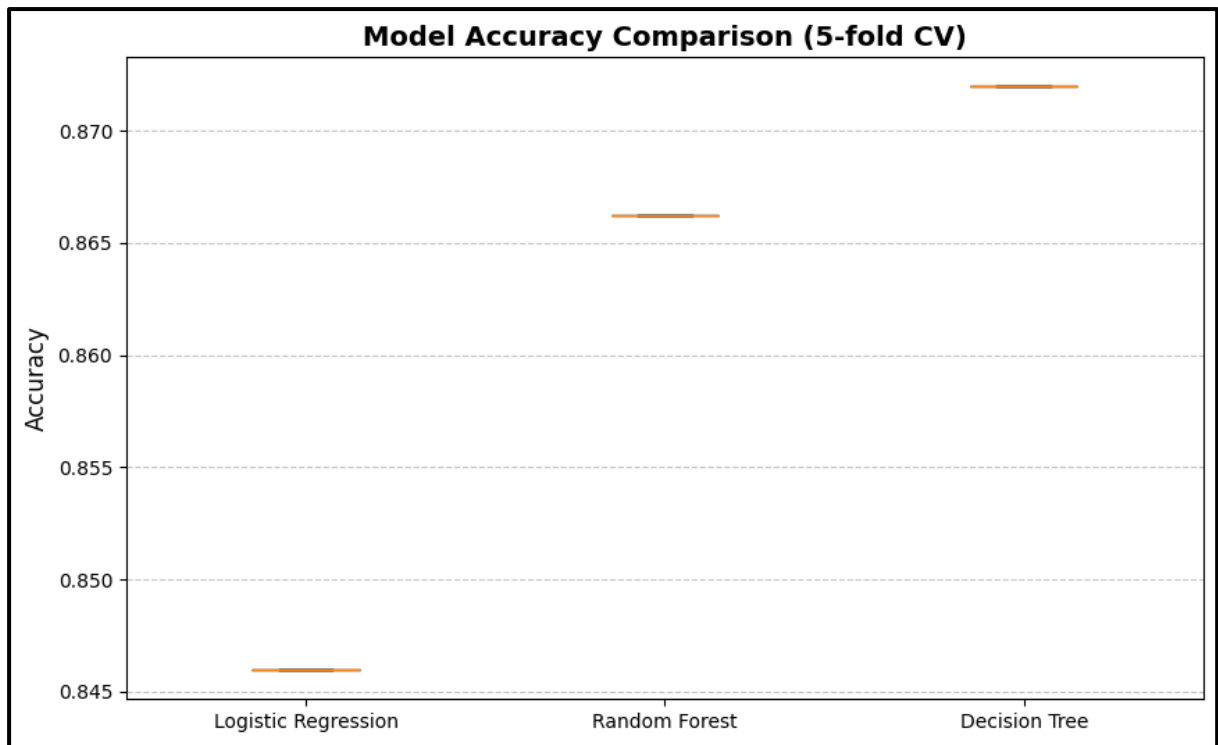


Figure 7: **Box plot of model accuracies (binary classification)**

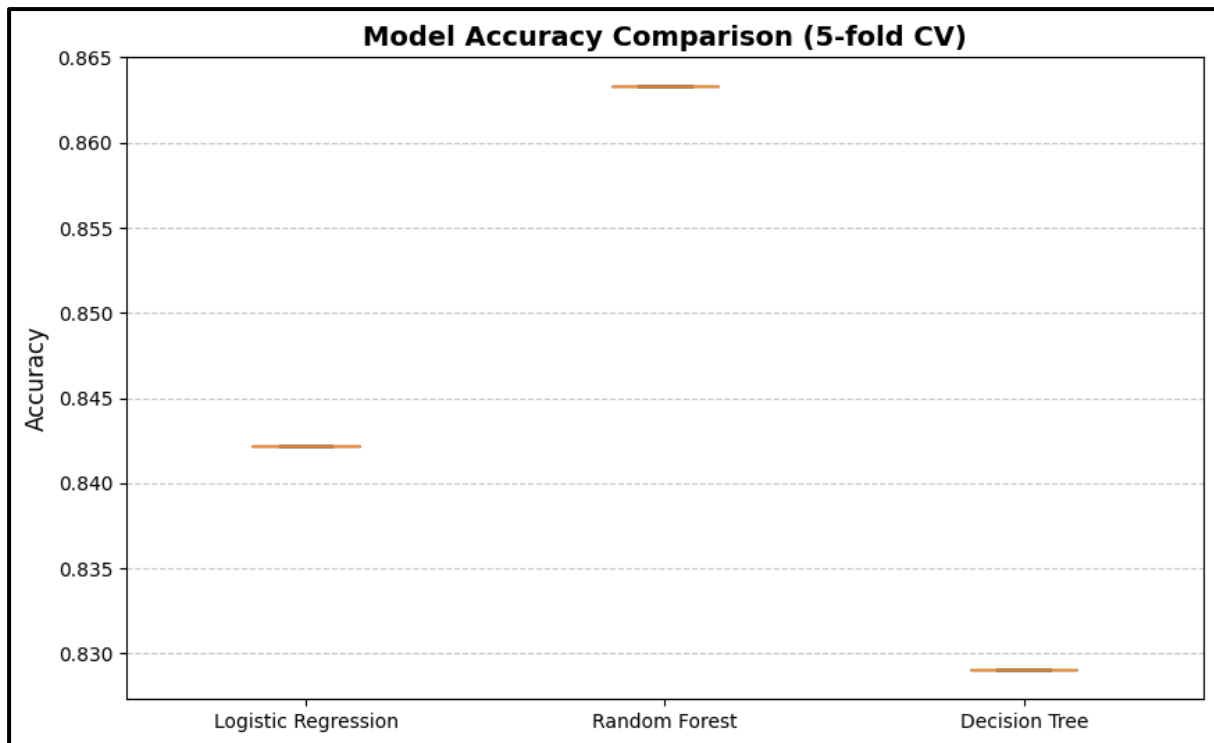


Figure 8: Box plot of model accuracies (multiclass classification)

ROC-AUC Curves

The ROC-AUC scores revealed that Random Forest and Decision Tree achieved values above 0.85, showing superior classification ability. Logistic Regression performed slightly lower, particularly in rare classes. ROC plots visually demonstrated the robustness of ensemble models in distinguishing normal from attack traffic.

Visualizations:

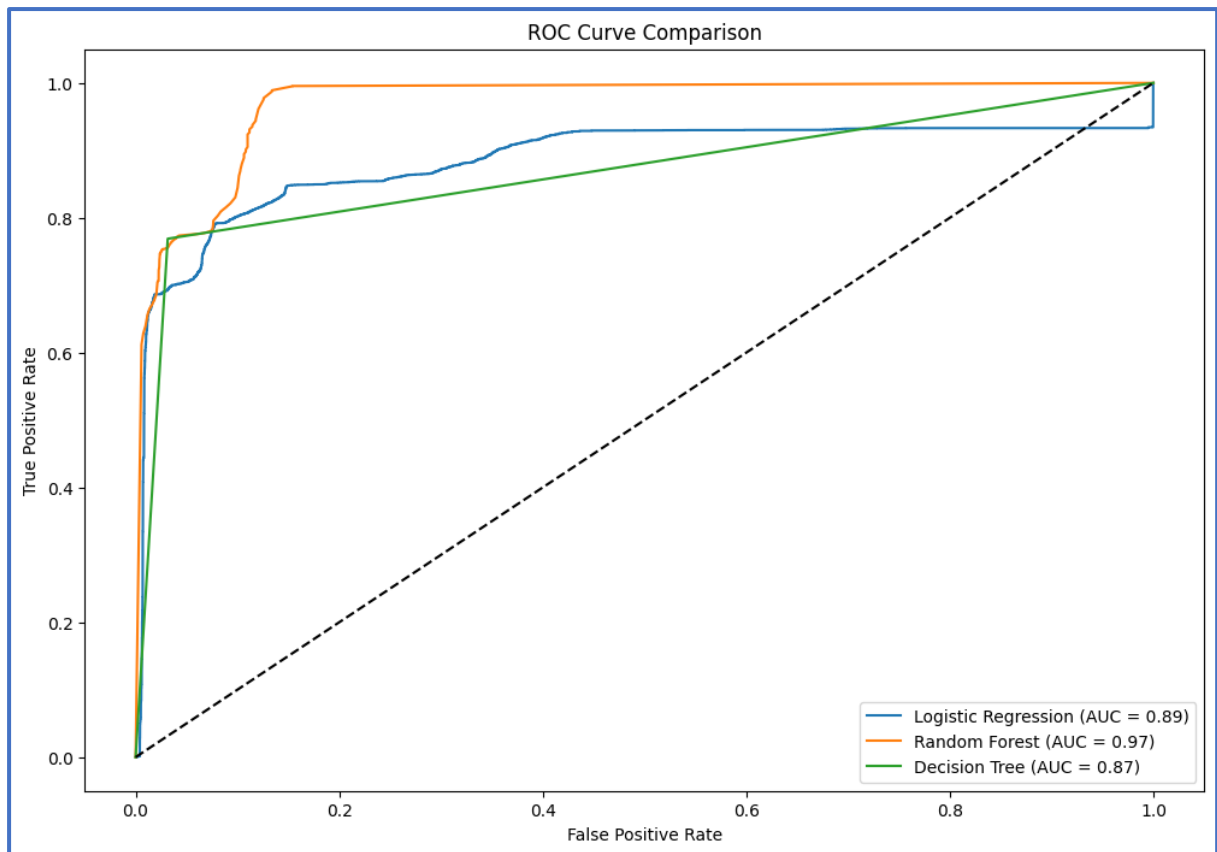


Figure 9: ROC-AUC curves for binary classification.

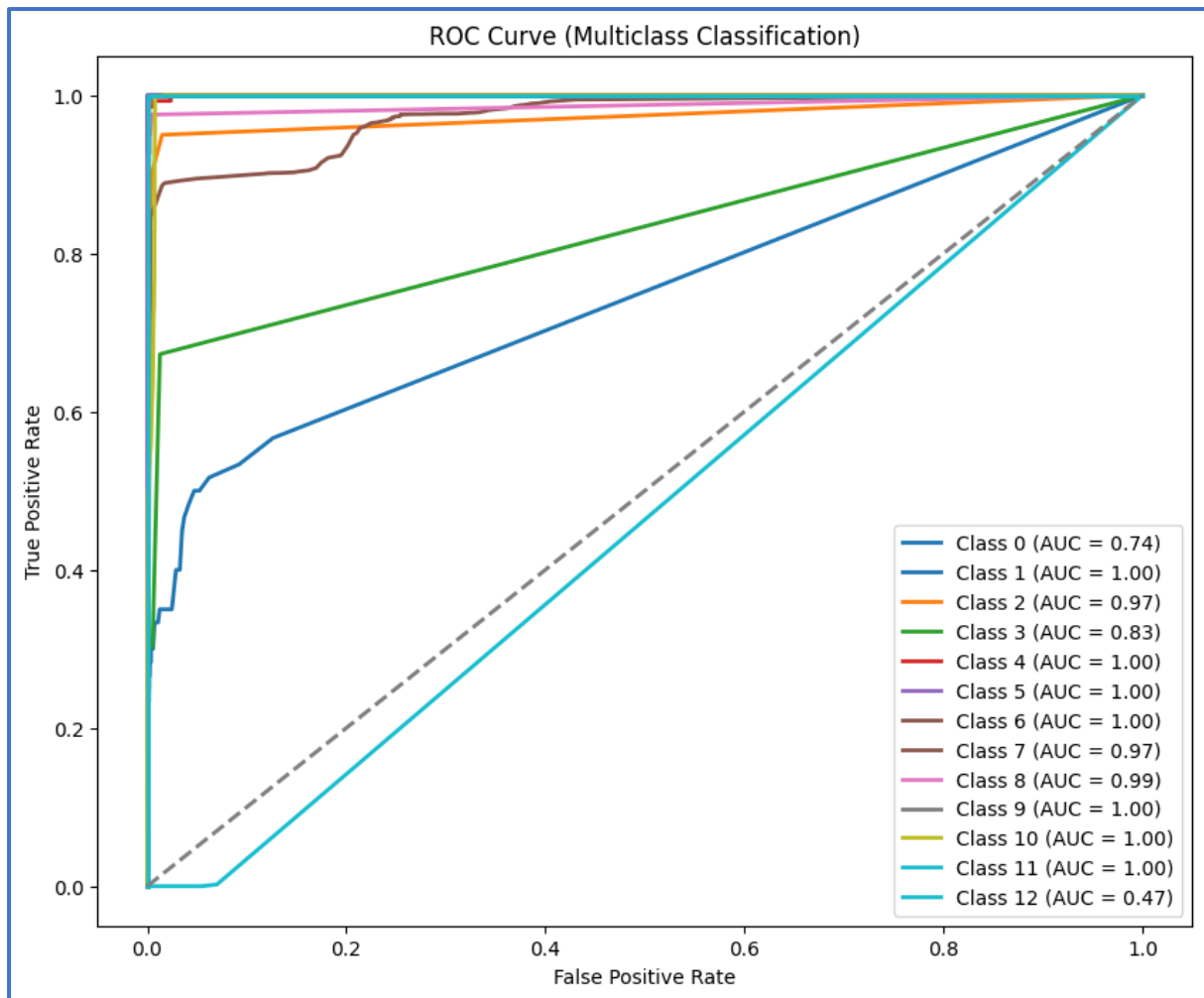


Figure 10: ROC-AUC curves for multiclass classification.

Discussion

The experiments revealed that Logistic Regression, while efficient, failed to detect rare classes reliably. Random Forest and Decision Tree showed better balance between accuracy and fairness across categories. The SMOTE-Tomek approach contributed significantly to improving minority class recognition.

Evaluation using macro F1-scores ensured that minority classes were not overshadowed by dominant categories. The findings suggest that ensemble models are better suited for IDS applications in highly imbalanced datasets.

Conclusion

This project successfully implemented machine learning techniques for intrusion detection using the NSL-KDD dataset. Preprocessing, hybrid resampling, and multiple classifiers led to meaningful results. Random Forest and Decision Tree emerged as the best-performing models, particularly in handling imbalanced classes.

The emphasis on macro F1-scores highlighted the importance of fair evaluation across all categories. The work demonstrates the potential of hybrid sampling and ensemble learning in enhancing IDS performance.

Future Works

Future work may focus on applying deep learning techniques such as CNNs and LSTMs to exploit temporal and spatial patterns in network traffic. Testing on larger datasets such as CIC-IDS-2017 or UNSW-NB15 will validate generalization.

Further improvements can include real-time deployment of the IDS and the use of ensemble deep learning methods. Feature selection and dimensionality reduction may also improve interpretability and computational efficiency.